

Intro to Linear Mixed Models: Modeling continuous longitudinal outcomes

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Some motivational datasets

Before we start, I would like to introduce some datasets that will set the scene for this session (correlated continuous outcomes) and the next session (correlated categorical outcomes). All three datasets come from R libraries used for longitudinal data modeling:

- 1 Sleep study (`lme4`)
- 2 Autism data (`WWGbook`)
- 3 Respiratory data (`geepack`)

The first two will be used to demonstrate linear mixed models (LMMs), and the third dataset (which has a binary outcome) will be used in the next session to consider Generalized Linear Mixed Models (GLMMs) and Generalized Estimating Equations (GEEs).

Study 1: Sleep data

This study, conducted by Belensky(2003), investigates the effect of sleep deprivation on reaction times. On day 0 subjects had normal amounts of sleep, and starting that night, had 3 hours sleep per night. Average reaction time was measured across a battery of tests administered each day, to each subject. Variables are:

- **Reaction:** Average reaction time (**continuous outcome**)
- **Day:** Number of days of sleep deprivation (day: 0,1,2,...,9)
- **Subject:** Subject ID

Questions for later:

- 1 Which **factors** is **FIXED** and which is **RANDOM**?
- 2 Is the single fixed effect a **WITHIN**-subject, or **BETWEEN**-subject effect?

Study 2: Autism data

This study (Oti *et al*, 2006) investigates the effect of the level of communication development (classified at age 2) on social development in Autistic children. All cohort subjects are initially measured at 2 years old and then followed up until age of 13:

- **VSAE**: parent-reported Vineland Socialization Age Equivalent
- **Age**: Age in years (2, 3, 5, 9, 13)
- **Sicdegp**: Expressive language score at 2yo:Low, Med, High
- **Childid**: Unique child identifier

Now:

- 1 Of the three factors, which are Fixed and which is Random?
- 2 For each fixed factor: **Within-** or **Between-** subject?

Study 3: Respiratory data(for next session)

A multi-centre, placebo-controlled RCT to investigate the efficacy of a 'drug' on respiratory illness. A group of 111 patients (from two centres) were randomized to either the placebo or treatment arm. Respiratory illness (y/n) was observed at baseline, and then again on three subsequent visits. Variables:

- **Respiratory illness** present (y/n);
- **Visit:** 1 (baseline) and three follow-up visits (2,3 and 4);
- **Treat:** P=Placebo or A=Active
- **Patient.id:** Unique patient identifier
- **Centre.id:** Centre ID (1 or 2)

Respiratory data(...next sessio)

Now for respiratory data:

- 1 How does the outcome differ (compared to Sleep and Autism studies)?
- 2 Of the four **EFFECTS**, which are **FIXED** and which are **RANDOM**?
- 3 For each **FIXED** effect, which is **WITHIN**- and which is **BETWEEN**-subject

BONUS POINTS: What is the name of the rather simplistic 'classical' bivariate tests that has been traditionally used for binary outcomes for pre-post studies (similar to this)? *Hint: Only works for 2×2 data*

Rationale:

Q: Why did I introduce these datasets before I have provided you the appropriate tools to analyze them? What features do all of them share?

A: Repeated measurement, or more generally, correlated outcomes. More soon.

So what will we cover?

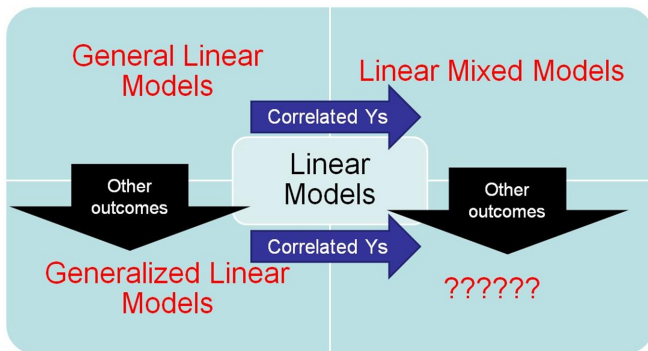
Using models you should already be familiar with (linear regression, the general linear model and the generalized linear model), we will extend these models to deal with (and make use of) correlated clinical outcomes.

What we will cover....

- 1 Introduction
 - Longitudinal data
 - Fixed and Random effects
- 2 Methods for the analysis of (continuous) longitudinal data
 - 'Classical' approaches
 - Linear Mixed Models
 - Worked example of a Linear Mixed Model in R
- 3 Sample size calculations for LMMs

Overview of linear models

(Simplified) unifying theory of linear models



Repeated measures data

- The models we have cover up to this point are for the analysis of cross-sectional (uncorrelated) data
- What if our observational units are measured repeatedly?
- If repeated measurements are over time, such data are called **repeated measures**, **time-course**, or **longitudinal** data
- Studies involving such data are often called **longitudinal** studies (usually a cohort study, although RCTs can also be longitudinal)

Answer to previous question: Terminology

Take care of the term '**cohort design**'. By definition, it just means (measured) exposure precedes (measured) outcome, and such studies **MAY**, or **MAY NOT** involve repeated measurements of the outcome

Repeated measures data

What difference does it make?

- What's the harm if we just use our standard cross-sectional regression methods (general linear models, generalized linear models etc.)?
- What assumption do all these 'cross-sectional' methods share that would be violated?
- **ANS: Independence!!**
 - Each (X, Y) observation is assumed to be independent of every other (X, Y) observation
 - That is, (X_i, Y_i) is independent of (X_j, Y_j) ; for all i, j where $i \neq j$

Repeated measures data

Still....So what??

- What's the problem with 'correlated' observations?
- We can assume measurements taken for a single subject (i.e. WITHIN-subject) are likely to be more similar than observations among subjects (BETWEEN-subjects)
- Lower variation for within-subject observations should be considered the advantage it is (if modelled properly)
- BUT problem with cross-sectional methods is that they assume all variation is 'between-subject' (so can underestimate variation in the data)
- SO, if we use a standard cross sectional approach, variance estimates will be under-estimates $\Rightarrow$ type I errors... **We are not 'telling' our model that the data are repeated**

Think about this. WHY?

Advantages of repeated measures data

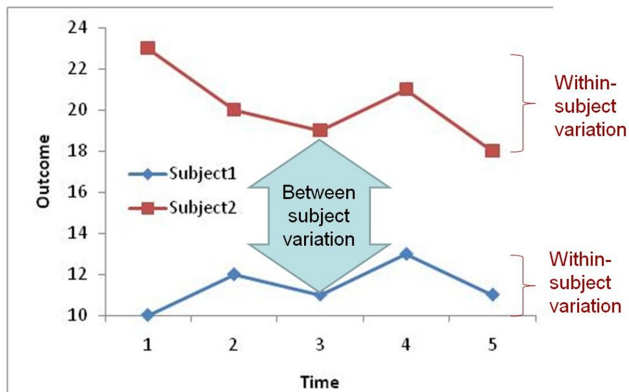
- Remember that we are trying to reduce error and minimize bias (E.g. adjusting for other covariates)
- In repeated measures analyses, individuals are considered as 'blocks' of observations
- Assume within-subject variation lower than among subjects (between-subject variation)
- Advantage - can conduct complex designs with fewer observational (or experimental) units

Between vs Within subject variation

- Between-subject (*between-patient*) variation is how we would expect differences between subjects (or groups thereof) to manifest itself (this is the type of variation we are used to) - e.g. A standard parallel RCT
- Within-subject variation is how a **given** individual varies over time (or over space: e.g. left eye vs right eye)

Between vs Within subject variation

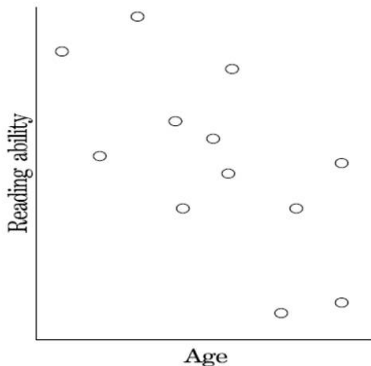
Figure: Within Vs Between variation for two individuals



Repeated measures data

Not accounting for repeated measurement

- Cross-sectional data (observations are independent)
- Greater age $\Rightarrow$ Poorer ability

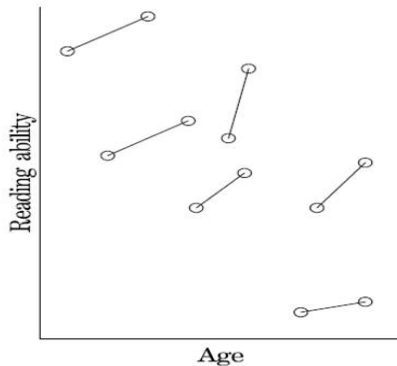


Source: Diggle et al., Analysis of Longitudinal Data

Repeated measures data

Accounting for repeated measurement

- Longitudinal data (observations are linked)
- Greater age $\Rightarrow$ Better ability



Source: Diggle et al., Analysis of Longitudinal Data

Other types of correlated data

Clustered data

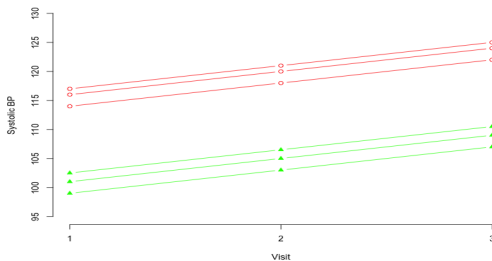
- We can think of longitudinal data as repeated measures over time.
- Clustered data is often repeated measures over space
 - E.g. If we consider a multi-centre study involving a number of clinics we would expect patients attending same clinics to be more similar (relative to those attending other clinics).
 - We need to account for this in modelling
- Good news is that methods used for clustered data are identical to those used for longitudinal data
- Clustered data turns up in health all the time. E.G.
 - Ophthalmology: Right and left eyes of the same patient
 - Multi-center clinical trials and cohort studies

Within-subject correlation

- Between-variable correlation considers the association between two variables (e.g. Exercise and depression levels) - correlation we are used to thinking about (e.g. $r = 0.7$)
- Within-subject correlation considers the correlations between observations at two times (for each subject)
 - For example, White blood cell count before and after a treatment
 - Those with higher before-treatment WBC would be expected to have higher WBC after treatment (relative to other subjects)
- As you would expect, within-subject correlation is often positive (patients who start higher, end higher)

The trade off: Independent information

Intuitively, we might expect the more correlated our observations are, the fewer patients needed in a study. But actually, the opposite is true. Higher levels of within-patient correlation implies we need a LARGER sample size. Consider the following case with very high within-subject correlation ($r_{ICC} = 0.95$):



Do we really need to follow-up to say groups are different? No the extra repeated measures give very little new information.

Random effects: "ID" variables

- **Random effects** are those where the particular '**groups**' of **observations** (levels) are drawn **randomly** from a population of groups.
- For example, if we have 10 patients drawn from a population of patients on which we take five (repeated) measurements.
- It doesn't make sense to test:
$$H_0 : \mu_{subject1} = \mu_{subject2} = \dots = \mu_{subject10}$$
- If we rejected this hypothesis, we can't make any meaningful inference about the differences between the specific patients (to the population)
- If we were going to tests a hypothesis about this random effect (Patient), which we rarely do, we would test:
$$H_0 : \sigma^2_{patients} = 0 \text{ (patients in popn. don't vary)}$$
- If a variable has **"ID"** in the name (e.g. patient_ID) it is almost always a **random effect**

Your turn: Understanding fixed and random effects?

Fixed effects are the ones we know (and love), where we would expect a differences between two groups to be **FIXED** as we move from the sample to the population

- Which (below) are **Fixed**- and which **Random**-effects??
- **AND** for fixed effects (only), which are **within-subject** and which **between-subject**?
 - ① New treatment vs Placebo-control (in parallel RCT)
 - ② Patient (A, B, C, D, E, F or G)
 - ③ Visit (Baseline, 3 month, 6 month and 1 year)
 - ④ Type of Cancer (Lung, Colo-rectal, Liver, Pancreatic)
 - ⑤ Brand of toothpaste (A, B or C) → **Tricky!!!**

What about datasets at beginning of lecture??

Fixed and random effects

- Generally only hypotheses about fixed effects interest us.
- **So why consider random effects at all?**
Ans: Because we want to (a) deal with the **problems** of correlated outcomes (*nuisance effect*); and (b) take advantage of the **additional information** repeated measurement yields.
- Problem with random effects is we often need a lot of model parameters (i.e. 40 subjects $\geq$ 40 parameters)
- Now we understand the purpose of fixed and random effects, we can start to consider the different methods (models) for the analysis of repeated measure data

Methods for longitudinal continuous outcomes

We will discuss four methods for the analysis of continuous longitudinal outcomes:

- 1 Repeated measures ANOVA (RM-ANOVA)
- 2 Repeated measures multivariate ANOVA (RM-MANOVA)
- 3 Linear mixed models (LMM)
- 4 Linear marginal models (Next session)

We will only cover the last two of these in any detail

Classical approaches

- The first two of these techniques have fallen out of favour recently (as they make restrictive, and often, unrealistic assumptions about the data, or they are not practical)
- But I will cover them briefly
- Why bother? They illustrate useful insights into repeated measures data and its modelling
- They are also still widely used in some branches of health science (e.g. exercise and rehabilitation sciences)
- Also (for interests sake) these models can be mimicked (very closely) in a Linear marginal model anyway (so we can directly compare and assess their adequacy) → More next session

Assumptions of RM-ANOVA

Beyond those assumptions normally associated with an ANOVA, the RM-ANOVA makes more:

Compound symmetry

- 1 Constant variance $\rightarrow \sigma^2(Y_{ij}) = \sigma_e^2 + \sigma_b^2$
- 2 Constant covariance $\rightarrow \text{Cov}(Y_{ij}, Y_{ik})$

The second of these implies that regardless of how far apart two repeated measures are, they will be just as correlated
e.g $\text{Corr}(\text{Day1}, \text{Day2}) = \text{Corr}(\text{Day1}, \text{Day200})$

These assumptions are captured in the statement of **Compound symmetry**. Compound symmetry can be represented as a type of **residual (or error) covariance structure** (We cover covariance structures in MUCH more detail when we get to Marginal Models).

Why not RM-ANOVA?

- The compound symmetry (covariance structure) assumption associated with the Repeated measures ANOVA renders it too unrealistic in many health-based longitudinal studies.
- Besides, many stats packages (esp. SPSS) tend to make the use of RM-ANOVAs quite painful
- We will only implement the RM-ANOVA by using the much more flexible Linear Marginal Model (next session)
 - i.e. RM-ANOVA $\rightarrow$ Linear Marginal Model with Compound symmetry residual covariance structure (**more later.**)

Repeated measure MANOVA

- RM-MANOVA (Repeated Measures **Multivariate** Analysis of variance) represents an alternative approach to the analysis of repeated measures data
- More computationally intensive than RM-ANOVA, but it relaxes some of the (often unrealistic) assumptions of the RM-ANOVA in regards to the residual covariance structure (no compound symmetry).
- In the standard MANOVA approach, no particular structure is assumed in the residual covariance structure (i.e. The covariance structure is called **unstructured**)
- Again, we can implement the RM-MANOVA using the Linear Marginal Model
 - i.e. RM-MANOVA $\approx$ Linear Marginal Model with Unstructured residual covariance structure (**more later.**)

Why we won't use classical approaches

- Generally, RM-ANOVAs and RM-MANOVAs present major restrictions that make their use impractical.
- RM-ANOVAs assumptions about a compound symmetric covariance structure are too simplistic (i.e. often invalid) in most health studies.
- RM-MANOVA needs a large sample sizes in order to be useful, also has a major problem with missing values
- Both methods have major problems with:
 - Different numbers of observations for different subjects
 - Variation in timing of individual subject measurement (e.g. Subject 1: Days 1, 10, 20: Subject 2: Days 2, 11, 19 etc)
 - If even one value is missing for a patient, that patient must be excluded (list-wise deletion $\Rightarrow$ complete-case bias?)

Linear Mixed Models for longitudinal data

This leaves the last two methods we will talk about:

- ① Linear **MIXED** models mix (consider) both **fixed** and **random** effects (hence the name 'Mixed' models)
- ② Linear **MARGINAL** models represent an alternative approach accounting for correlation of observations through the use of **Residual covariance structures**

Keeping it simple

While covering these comparatively complex models, remember one thing: **These models are a way of dealing with correlated data so we can consider research questions in much the same way as independent data; Inferences about within-subject effects can be conducted in a similar way to (standard) between-subject effects**

Linear Mixed Model: A general representation

The Linear Mixed (Effects) model can be represented:

$$Y_{ti} = \beta_0 + \beta_1 X_{ti}^{(1)} + \beta_2 X_{ti}^{(2)} + \cdots + \beta_{k-1} X_{ti}^{(k-1)} \\ + u_{0i} + u_{1i} Z_{ti}^{(1)} + \cdots + u_{qi} Z_{ti}^{(q)} + \epsilon_{ti} \quad (1)$$

The value of t ($t = 1, 2, \dots, n_i$) indexes the longitudinal observations for each subject i ($i = 1, 2, \dots, m$) over time. Using matrix notation:

$$Y_i = X_i \beta + Z_i u_i + \epsilon_i$$

The first part of the model, containing $X_i \beta$, is the **fixed effect(s)** component of the model (a standard general linear model), whereas the $Z_i u_i$, represents the **random effect(s)**.

LMMs: Defining the terms

In the model:

$$Y_i = X_i\beta + Z_iu_i + \epsilon_i$$

Y_i is a vector of continuous responses for the i^{th} subject

X_i is an $n_i \times k$ design matrix for the corresponding fixed effect covariate values for each of the n_i observations collected on the i^{th} subject

β is a vector of regression coefficients for the fixed effects, X

Z_i is the $n_i \times q$ design matrix that represents the known values of the q covariates, $Z^{(1)}, \dots, Z^{(q)}$, for the i^{th} subject

u_i is a vector of q random effects (e.g. patient effect), with $u_i \sim N(0, D)$

and ϵ_i is the residual, with $\epsilon_i \sim N(0, \sigma^2 I_n)$

Next we will define the matrices, D

The matrix D

- The D matrix represents the variance-covariance matrix of the random effects
- Elements along the main diagonal of the D matrix represent the variances of each random effect in u_i , and the off-diagonal elements represent the covariances between two corresponding random effects
- We will talk more of D when we look at various LMMs
- In particular, this matrix is used to define the random intercept and random coefficient models, but we rarely interpret it.

z and D

The elements D really just specify the relationships of the random effects (e.g. patients) to our outcome variable. The coefficients estimated in D rarely interest us.

Keeping it simple: Linear models, Mixed Models and Marginal models

A standard general linear model:

$$y_{ij} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \epsilon_{ij}$$

To deal with the correlated data, the linear mixed model and linear marginal models...

$$y_{ij} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \text{Put something here}$$

The 'something' is quite different between the **mixed** model approach (which is a 'subject-specific' or **conditional** model), and the **marginal** model approach (a **population-averaged** approach)

Keeping it simple: Mixed vs Marginal models

For **mixed** (aka conditional, or subject-specific) models, the correlated data is dealt with EXPLICITLY in the model (i.e. terms are added to deal with random effects)

$$y_{ij} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \text{random effects} + \epsilon_{ij}$$

Each patient has their own model (ie. subject-specific model)

IN CONTRAST, for **marginal** (aka population-averaged) models, the correlated nature of the data is dealt with in the residual term ϵ

$$y_{ij} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \epsilon_{ij}$$

where $\epsilon_{ij} \sim N(0, R)$ where R is the **specified** residual covariance structure (and assumes R **FULLY** reflects correlations among observation); Everyone SHARES this structure.

Our first LMM: Random intercept model

- Let's start with a quite simple LMM, the **Random intercept model**
- This model allows each subject(after accounting for the fixed effects) to vary at baseline
- But we should note that the effect of time, is assumed to be the same for all individuals (all individuals share a common slope over time)
- SO patients get their own intercept, but share a slope (over time)

Mixed models are subject-specific models

The reason mixed models are often called **subject-specific models** is that each patient effectively gets their **own model**
⇒ each patient has their **own regression line** BTW This is why I like them better than any other model

Random intercept model specification

Taking the (general) linear mixed model specified above:

$$Y_i = X_i\beta + Z_iu_i + \epsilon_i$$

In the random intercept model, the Z matrix (as there are only intercepts) becomes a column of ones, so the model simplifies down to:

$$y_{ij} = X_{ij}\beta + u_i + \epsilon_{ij}$$

Now defining the terms.....

Our first LMM: Random intercept model

The Random intercept model:

$$y_{ij} = X_{ij}\beta + u_i + \epsilon_{ij}$$

Where

y_{ij} is the the j^{th} repeated measure on the i^{th} subject

X_{ij} is the corresponding vector of fixed effect covariates/factors

β is the corresponding fixed-effect regression parameters (up to this point, a standard general linear model)

u_i is the (random) effect due to subject; The change (from average) intercept is subject i

ϵ_{ij} is the residual (error not explained by covariates); with

$$\epsilon_{ij} \sim N(0, \sigma_e^2)$$

$u_i \sim N(0, \sigma_u^2)$ is the variation among subjects not accounted for by covariates in X (in a random slope model this is a single coefficient, σ_i representing the variance of individuals from the mean intercept)

Simplifying the random intercept model

If we do a little algebraic rearrangement:

$$y_{ij} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + u_i + \epsilon_{ij}$$

Can be rearranged to:

$$y_{ij} = (\beta_0 + u_i) + \beta_1 X_1 + \beta_2 X_2 + \cdots + \epsilon_{ij}$$

So we can see u_i is the **difference from the average intercept, β_0 , due to patient i**

Sleep data

Recall Belensky(2003) sleep study. We want to investigate the effect of sleep deprivation on reaction times. On day 0 subjects had a normal amount of sleep, and starting that night, had 3 hours sleep per night. Average reaction time was recorded each day, for each subject. Variables are:

- **Reaction:** Average reaction time (**continuous outcome**)
- **Day:** Number of days of sleep deprivation (day: 0,1,2,...,9)
- **Subject:** Subject ID

We will start by running a **Random Intercept** Linear Mixed Model

Step 1: State hypotheses

$$H_o: \beta_{Days} = 0$$

(Reaction time is independent of amount of sleep deprivation)

$$H_a: \beta_{Days} \neq 0$$

(Amount of sleep deprivation does effect reaction time)

How does this differ from a standard linear regression hypothesis??

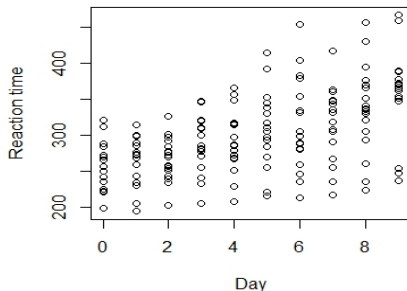
ANS:????

Note: I am assuming 'Day' is a continuous variable, I could have specified it as a factor (but 10 'groups' is too much)

*Also note for this problem we **only** have a within-subject fixed effect (we are not comparing groups of patients e.g. M vs F)*

Step 2: Eyeball the data

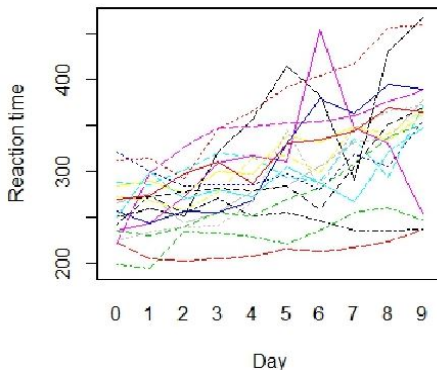
Let's start with a standard scatter plot (where there is no linking of within-subject observations)



Helpful?? Not really.

Step 2: Eyeball the data

Now let's link the within-subject observations



Much better. Note for later. Do you think the subjects have the same slope (i.e. the same response to sleep deprivation)?

Step 3: Specify the model

Now let's state the model in terms of our problem:

$$Reaction_{ij} = \beta_0 + \beta_{Day} Day + Subject_i + \epsilon_{ij}$$

Rearranging:

$$Reaction_{ij} = (\beta_0 + Subject_i) + \beta_{Day} Day + \epsilon_{ij}$$

$Reaction_{ij}$ is the reaction time of the i th subject on the j th day

β_0 is the overall (average) y-intercept

$Subject_i$ is the CHANGE from β_0 due to being subject i

β_{Day} is the slope due to day (expected change in reaction time due to an extra day of sleep deprivation)

ϵ_{ij} is the random (unexplained) error associated with $Reaction_{ij}$

Step 4: Run the model

- Several libraries in R run LMMs. I prefer `lme4` library
- This doesn't come with the base R (need to download)
- Assume our data already read in and stored in `sleep.df`

R syntax: Random intercept LMM

```
library(lme4)

# Need a null (nothing in it) model first
null.mod<-lmer(Reaction ~ 1 + (1|Subject), data=sleep.df)

# Now fit our model
rint.mod<-lmer(Reaction ~ Days + (1|Subject), data=sleep.df)

# Does model represent an improvement over null
anova(null.mod, rint.mod)

summary(rint.mod)
```

Output 1a) ANOVA

```
> rint.mod<-lmer(Reaction~Days + (1|Subject), data=sleep.df)
> anova(null.mod, rint.mod)
Data: sleep.df
Models:
null.mod: Reaction ~ 1 + (1 | Subject)
rint.mod: Reaction ~ Days + (1 | Subject)

      Df    AIC    BIC logLik  Chisq Chi Df Pr(>Chisq)
null.mod  3 1916.6 1926.2 -955.28
rint.mod  4 1802.1 1814.9 -897.05 116.47      1 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

Output 1b) Coefficients

```
> summary(rint.mod)
Linear mixed model fit by REML
Formula: Reaction ~ Days + (1 | Subject)
Data: sleep.df
   AIC   BIC logLik deviance REMLdev
1794 1807 -893.2    1794    1786
Random effects:
Groups   Name             Variance Std.Dev.
Subject (Intercept) 1378.18   37.124
Residual                960.46   30.991
Number of obs: 180, groups: Subject, 18

Fixed effects:
              Estimate Std. Error t value
(Intercept) 251.4051    9.7459   25.80
Days         10.4673    0.8042   13.02
```

Random intercepts model: Interpretation

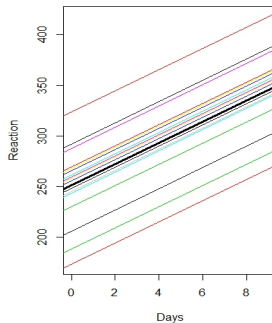
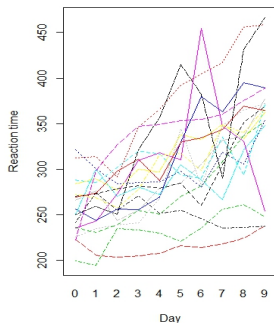
- We can see from output 1a that when add the fixed effect *Day* to the null model, there is a significant improvement ($\chi^2_{LR} = 116.47$, $df = 1$, $p < 0.001$) $\Rightarrow$ Significant effect of *Day* on reaction time
- When we go to the coefficients table (output 1b) we see $\beta_{Day} = 10.46$, that is, for every extra day of sleep deprivation, we expect reaction time to $\uparrow$ by 10.46 ms
- You will notice there is no R^2 (this is not a linear regression). Instead we are left with the MUCH LESS friendly **AIC = 1802.1**
- The AIC for the null model (see output 1a) is **AIC = 1910**, that our random intercept model is considerably lower suggest our model is a considerably better fit (remember **lower AIC means better model**)

Random intercepts model: Interpretation

- The average intercept was $\beta_0 = 251.4$ (see Output 1b)
- This implies the AVERAGE reaction time at baseline (after no sleep deprivation) was 251.4 ms
- Our **Random intercept** model allows individuals to vary from this average intercept
- Let's look at how this intercept varied across subjects...

Random intercept model: Adequacy

BUT does this model represent a realistic fit of the data?



Real data

Now, can you appreciate why mixed models are called **subject-specific models**??

Fit (Random intercepts)

Linear Mixed Model: Random coefficients model

- Perusing the graphs on the previous page suggests that the model doesn't fit the data well.
- It appears the assumption that all subjects have the same slope (the same [average] response to progressive sleep deprivation) is unrealistic
- A better model may be the **random coefficients model**. This allows subjects to have BOTH their own intercept, and their own slope...

Back to Step 3: Model (Re)specification

The random coefficients model (in our case), will be

$$Reaction_{ij} = \beta_0 + \beta_{Day} Day + Subject_{i0} + Subject_{Days,i} Days + \epsilon_{ij}$$

Rearranging:

$$Reaction_{ij} = (\beta_0 + Subject_{0i}) + (\beta_{Day} + Subject_{Days,i}) Day + \epsilon_{ij}$$

WHERE HAVE YOU SEEN SOMETHING LIKE THIS BEFORE??

Let's define each of the terms

Back to Step 3: Model (Re)specification

$$Reaction_{ij} = (\beta_0 + Subject_{0i}) + (\beta_{Day} + Subject_{Day,i})Day + \epsilon_{ij}$$

$Reaction_{ij}$ as before

β_0 is the overall (average) y-intercept

$Subject_{0i}$ is the change in y-intercept due to subject i

β_{Day} is overall (average) slope for day

$Subject_{Day,i}$ is change in slope due to subject i

So we can see $Subject_{Day,i}$ as the subject 'effect modification' of days of sleep deprivation associated with subject i

Step 3 Model (re)specification: Random coefficients model

R syntax: Random coefficient LMM

```
# Fit random coefficients model
rcoef.mod<-lmer(Reaction ~ Days + (Days|Subject),
data=sleep.df)
# Is it an improvement on the random intercepts model)
anova(rint.mod, rcoef.mod)

summary(rcoef.mod)
```

Output 2a) ANOVA

```
> rcoef.mod<-lmer(Reaction~Days + (Days|Subject), data=sleep.df)
> #Significant improvment over random intercept?
> anova(ran.int, ran.coef)
Data: sleep.df
Models:
ran.int: Reaction ~ Days + (1 | Subject)
ran.coef: Reaction ~ Days + (Days | Subject)
      Df    AIC    BIC logLik Chisq Chi Df Pr(>Chisq)
ran.int   4 1802.1 1814.9 -897.05
ran.coef   6 1764.0 1783.1 -875.99 42.113      2 7.164e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

Output 2b) Coefficients

```
> summary(ran.coef)
Linear mixed model fit by REML
Formula: Reaction ~ Days + (Days | Subject)
Data: sleep.df
AIC   BIC logLik deviance REMLdev
1756 1775 -871.8    1752    1744
Random effects:
Groups   Name             Variance Std.Dev. Corr
Subject  (Intercept)  612.092  24.7405
          Days        35.072   5.9221  0.066
Residual                654.941  25.5918
Number of obs: 180, groups: Subject, 18

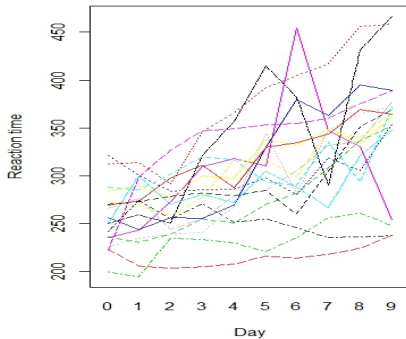
Fixed effects:
              Estimate Std. Error t value
(Intercept)   251.405     6.825    36.84
Days           10.467     1.546     6.77
```

Step 5: Interpreting results

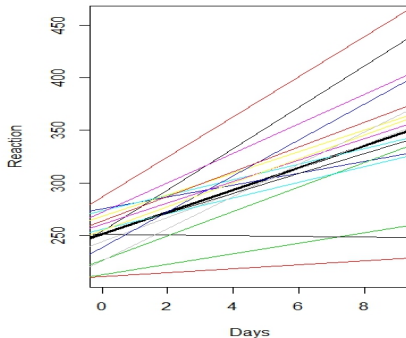
- First we should note that the random coefficients model represents a significant improvement over the random intercept model ($\chi^2_{LR} = 42.113$, $df = 2$, $p < 0.001$)
- BUT is the extra model complexity worth it. Remember AIC is a parsimony measure which accounts for **FIT and COMPLEXITY** (The LR statistic is ONLY about fit)
- $AIC_{ranint} = 1802.1$ and $AIC_{rancoef} = 1764$. As random coefficient is lower $\Rightarrow$ a better model (worth it)
- Finally we should note that the fixed effect part of the model is exactly the same ($\beta_0 = 251.4$ and $\beta_{Day} = 10.46$) in both models (Note for β_{Day} , $t = 6.77$, $p < 0.05$)
- Finally, let's visualize this model to see if it seems a better fit to the data

Random coefficients model: Adequacy

MUCH BETTER. Hint: don't let your eyes get drawn too much by the LHS plot. Joining points is not really an 'accurate model'. *Think of a 'smoothed' line for each patient.*



Real data



Fit (Random coefficients)

Sample size calculation for LMMs (FYI)

- Like other methods, we can calculate sample size for Linear mixed models
- In fact the same formula can be applied to other models used for longitudinal continuous outcomes
- BUT the calculation I am about to show you, only applies when we have a single **Between-subject effect** and a single **Within-subject effect** (e.g. a longitudinal RCT)
- For more complex models/designs we have to go to the rule of thumb methods used for Linear Regression (and General Linear Models)

There is one additional quantity we need in a LMM sample size calculation: **The within-subject correlation, ρ** . This can be estimated from pilot data using the *ICC*, or 'guestimated'

Within-subject correlation

Within-subject correlations is something we go into detail about when we get to Linear Marginal Models, but in the meantime, I will give a rather simple definition:

Within-subject correlation is the extent to which measures taken from a single subject are correlated (i.e. How WELL can we predict the future from the past?)

For example, if my White blood cell count (compared to you) is high on day 1, it is likely to be high (compared to you) on day 2, etc...

Sample size formula for LMMs

The formula to calculate the sample size for an LMM (one within- and one between-subject effect) is:

$$N_{pergroup} = \frac{2(Z_{\alpha/2} + Z_{\beta})^2(1 + [t - 1]\rho)}{t(\Delta_{MCD}/\sigma)^2}$$

where:

t is the number of repeated measurements (for $t = 1, 2, \dots, t$)

ρ is the level of within-subject correlation

$Z_{\alpha/2}$, Z_{β} , σ and Δ_{MCD} are defined as before (standard t-test formula).

Recall:

- $Z_{\alpha/2}$ and Z_{β} are from a standard Normal distribution and reflect significance level and power, respectively
- σ : the within-group standard deviation
- Δ_{MCD} is the minimal **CLINICAL** difference

Sample size for LMMs: Worked example

Let's consider the Autism data. Recall:

- The outcome is level of socialization (VSAE score)
- The **Within-subject** effect is **Age** (2,3,5,9,13yo)
- The **Between-subject** effect is **Sicdegp at 2yo** (language score): Low, Med, High

Note that we have **three** 'between-subjects' groups here (low medium and high), but we are used to thinking about the two group case. BUT all this means is that we have to multiply $N_{pergroup}$ by 3 (rather than 2)

Sample size for LMMs: Worked example

Ingredients

Now let's start collecting our *ingredients*:

- ① **Type 1 error:** $\alpha = 0.05$
- ② **Type 2 error;** $\beta = 0.1 \Rightarrow \text{Power} = 90\%$
- ③ **σ (Standard deviation):** Pilot data (backed by previous studies) suggest that we would expect to see a standard deviation (for a certain age, for a certain language group) of 15 units: $\sigma = 15$
- ④ **Δ_{MCD} (Minimal clinical difference):**
CONSULTATION WITH EXPERTS suggests that a difference in socialization score (VSAE) less than 10 would not represent a scientifically important difference:
 $\Delta_{MCD} = 10$

Now, these are the quantities we are used to, what about the quantities that are specific to repeated measures studies?

Sample size for LMMs: Worked example

Ingredients

Continuing...

- ⑤ **t**: In this study we are considering 5 repeated measurements (time points: 2, 3, 5, 9 and 13yo): **$t=5$**
- ⑥ **ρ** : Within-subject correlation. This can be a tricky one (particularly if we don't have pilot data). In the absence of knowledge, I will assume there is a moderate level of within-subject correlation: **$\rho = 0.6$**

For a rough guide (when you don't know)

- $\rho = 0.3 \Rightarrow$ Weak within-subject correlation
- $\rho = 0.6 \Rightarrow$ Moderate within-subject correlation
- $\rho = 0.9 \Rightarrow$ Strong within-subject correlation

Sample size for LMMs: Worked example

Calculation

Now we have all of the ingredients, let's perform the calculation:

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2(1 + [t - 1]\rho)}{t(\Delta_{MCD}/\sigma)^2}$$

$$N_{pergroup} = \frac{2(1.96 + 1.28)^2(1 + [5 - 1]0.6)}{5(10/15)^2}$$

$$N_{pergroup} = \frac{21(1 + 2.4)}{5(0.666)^2} = \frac{71.4}{2.222} = 32.13$$

So we need 33 children per group $\Rightarrow 3$ (language group) * 33 = 99 children (observational units).

$\Rightarrow$ 5 times points so we will have **N=5x99=495** observations.

Sample size for LMMs vs cross-sectional

Just out of interest, let see what we would get if we were doing this study cross sectionally (i.e. $\rho = 0$)

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2(1 + [t - 1]\rho)}{t(\Delta_{MCD}/\sigma)^2}$$

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2(1 + [t - 1]0)}{t(\Delta_{MCD}/\sigma)^2}$$

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2(1)}{1(\Delta_{MCD}/\sigma)^2}$$

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2}{\Delta_{MCD}^2/\sigma^2}$$

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2\sigma^2}{\Delta_{MCD}^2}$$

Recognize this formula???

Sample size for LMMs vs cross-sectional

Plugging in ingredients (1-4 only)

$$N_{pergroup} = \frac{2(Z_{0.025} + Z_{0.1})^2 \sigma^2}{MCD^2}$$

$$N_{pergroup} = \frac{2(10.5)15^2}{10^2} = \frac{21 * 225}{100} = 47.25$$

So for a cross-sectional study (comparing the three language groups) we would need $N = 3 * 48 = 144$ individuals (and 144 observations)

In terms of OBSERVATIONAL UNITS (patients), which is more efficient?

In term of OBSERVATIONS, which is more efficient?

Mixed models and multi-level modeling

- The mixed effect models I have specified here are quite simple;
- I have generally only considered a single random effect (e.g. patientID) meaning that the $Z_i u_i$ term and D matrix are not that complicated
- BUT it is possible to have several random effects (patientID, LesionID, HospitalID) and these effects can be **crossed** or **nested**
- Such models fall into the area of **MULTI-LEVEL MODELING**

Multi-level modeling

Multilevel study designs in the clinical sciences are usually not too complex (there are exceptions), but in the social sciences (e.g. political science) they can get quite complex. In these studies, the $Z_i u_i$ term and D matrix can get quite nasty.

Where to from here

- There is one more model I want to cover when it comes to modelling continuous longitudinal data: The Linear Marginal Model
- BUT I am going to leave this for next session. This is for two reasons:
 - ① Brain burn-out
 - ② Linear Marginal models are not used that often.
- So why cover them?? Because understanding Linear Marginal models gives a strong insight into how Generalized Estimating Equations (GEEs) work. GEEs are a generalized equivalent of the Linear Marginal model used for categorical correlated outcomes

THANK-YOU

Questions????